

# Linear Algebra – MAT 2610

Section 1.7 (Linear Independence)

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**Definition:** A set of vectors  $\{u_1, u_2, \dots, u_k\}$  is **linearly dependent** if there exist scalars  $c_1, c_2, \dots, c_k$ , not all zero, such that  $c_1u_1 + c_2u_2 + \dots + c_ku_k = 0$

**Example:** The set  $\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}, \begin{bmatrix} 3 \\ 5 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$  is linearly dependent since

$$2 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix} - 1 \begin{bmatrix} 3 \\ 5 \\ -2 \end{bmatrix} - 1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$



**Definition:** A set of vectors  $\{u_1, u_2, \dots, u_k\}$  is **linearly independent** if the only scalars  $c_1, c_2, \dots, c_k$  that satisfy  $c_1u_1 + c_2u_2 + \dots + c_ku_k = 0$  are  $c_1 = c_2 = \dots = c_k = 0$

**Example:** The set  $\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}, \begin{bmatrix} 3 \\ 5 \\ -2 \end{bmatrix} \right\}$  is linearly independent since the only solution to

$$a_1 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + a_2 \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix} + a_3 \begin{bmatrix} 3 \\ 5 \\ -2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

is  $a_1 = a_2 = a_3 = 0$



**Example:** Is  $\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix} \right\}$  linearly dependent or linearly independent?

$$x_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 0 \\ 2 & 5 & 0 \\ 3 & -1 & 0 \end{bmatrix} \xrightarrow{2r_1 - r_2 \rightarrow r_2} \begin{bmatrix} 1 & 2 & 0 \\ 0 & -1 & 0 \\ 3 & -1 & 0 \end{bmatrix}$$

$$\xrightarrow{3r_1 - r_3 \rightarrow r_3} \begin{bmatrix} 1 & 2 & 0 \\ 0 & -1 & 0 \\ 0 & 7 & 0 \end{bmatrix} \xrightarrow{-r_2 \rightarrow r_2} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 7 & 0 \end{bmatrix} \xrightarrow{\substack{r_1 - 2r_2 \rightarrow r_1 \\ r_3 - 7r_2 \rightarrow r_3}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$x_1 = 0$   
 $x_2 = 0$  | only sol'n, so lin independent



**Example:** Is  $\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 5 \end{bmatrix} \right\}$  linearly dependent or linearly independent?

$$x_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + x_3 \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}} + x_4 \begin{bmatrix} -1 \\ 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\left. \begin{array}{l} x_1 = 0 \\ x_2 = 0 \\ x_3 = 5 \\ x_4 = 0 \end{array} \right\} \text{solution}$$

linearly  
dependent



# Linear Independence of Matrix Columns

The columns of a matrix  $A$  are linearly independent if and only if the equation  $Ax = 0$  has only the trivial solution.



## Facts:

- The columns of a matrix  $A$  are linearly **independent** if the matrix equation  $Ax = 0$  has  $x = 0$  as its unique solution.
- The columns of a matrix  $A$  are linearly **dependent** if the matrix equations  $Ax = 0$  has an infinite number of solutions.
- The columns of a matrix  $A$  are linearly **independent** if the matrix equation  $Ax = b$  has at most one solution.
- The columns of a matrix  $A$  are linearly **dependent** if the matrix equation  $Ax = b$  has an infinite number of solutions.



**Example:** For what values of  $r$  is  $\begin{bmatrix} -2 & 1 & -1 \\ 0 & 1 & 1 \\ 1 & -3 & r \end{bmatrix}$  linearly dependent?

$$\begin{bmatrix} -2 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & -3 & r & 0 \end{bmatrix} \xrightarrow{r_1 + 2r_3 \rightarrow r_1} \begin{bmatrix} -2 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & -5 & 2r-1 & 0 \end{bmatrix} \xrightarrow{5r_2 + r_3 \rightarrow r_3} \begin{bmatrix} -2 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 4+2r & 0 \end{bmatrix}$$

↑      ↑      ↑  
basic basic ?

$x_3$  is free if and only if  $4+2r=0$ , or  $r=-2$

So, matrix lin dep if  $r=-2$   
lin indep if  $r \neq -2$





# Theorem 7

An indexed set  $S = \{v_1, \dots, v_n\}$  of two or more vectors is linearly dependent if and only if at least one of the vectors in  $S$  is a linear combination of the others. In fact, if  $S$  is linearly dependent and  $v_1 \neq 0$ , then some  $v_j$  (with  $j > 1$ ) is a linear combination of the preceding vectors  $v_1, \dots, v_{j-1}$ .

Ex: Say  $a_1 v_1 + a_2 v_2 + a_3 v_3 + a_4 v_4 = 0$   
if  $a_1 \neq 0, v_1 \neq 0$

$$a_1 v_1 = -a_2 v_2 - a_3 v_3 - a_4 v_4$$
$$\underbrace{v_1}_{\neq 0} = -\underbrace{\left(\frac{a_2}{a_1}\right)}_{\neq 0} \underbrace{v_2}_{\neq 0} - \underbrace{\left(\frac{a_3}{a_1}\right)}_{\neq 0} \underbrace{v_3}_{\neq 0} - \underbrace{\left(\frac{a_4}{a_1}\right)}_{\neq 0} \underbrace{v_4}_{\neq 0}$$



# Theorem 8

If a set contains more vectors than there are entries in each vector, then the set is linearly dependent. That is, any set  $\{v_1, \dots, v_p\}$  in  $\mathbb{R}^n$  is linearly dependent if  $p > n$

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 4 \\ -1 \\ 4 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} \right\}$$

4 vectors  
3 element in each vector ( $\mathbb{R}^3$ )  
 $\Rightarrow$  lin dep



# Theorem 9

If a set  $\{v_1, \dots, v_p\}$  in  $R^n$  contains the zero vector, then the set is linearly dependent.



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